Unit 1 – Using Objects and Methods
Lesson 2: Variables and Data Types



Keywords

- Keywords have a specific meaning to the compiler and cannot be used for anything else in the program.
- These are usually blue (or a special color) in your IDE.
- Common keywords that we will learn:
    - `public, private, protected`
    - `class, static, void, return, this`
    - `true, false, null`

Identifiers

- Identifiers are the names that identify the elements in a program. All identifiers must obey the following rules:
    - An identifier can only be _____
    - An identifier cannot start with a _____
    - An identifier cannot be a _____
- Descriptive identifiers make programs easier to read.
    - For example, using `x` as an identifier is okay, but `numberOfTeachers` is a more descriptive.
- Identifiers are case sensitive.
    - For example, `area`, `Area`, and `AREA` are three different identifiers.
- If the identifier consists of several words, we concatenate them into one, making the first word lowercase and capitalizing the first letter of each subsequent word (called camel case). For example: `dogName`, `dogAge`, `numberOfDogs`, etc.

Variables

- Variables are _____
- When we _____ a variable, we tell the computer to allocate some space in memory for something
- We tell it what the name of the thing will be (_____), so it knows how to find it later, and what the type of the thing will be (_____), so it knows how much space to set aside

How big of a space does it need? There are 8 _____ data types in Java and each of them need a different amount of memory space to store their values. Therefore, you have to declare the type first, followed by the identifier that it will be called.

Data Type	Range of Values it stores	Values	AP Exam
	–128 to 127	Integers	**Will only test on int
	–32768 to 32767		
	–2147483648 to 2147483647		
	–9223372036854775808 to 9223372036854775807		
	-3.4×10^{38} to 3.4×10^{38}	Decimals	**Will only test on double
	-1.8×10^{308} to 1.8×10^{308}		
	Thousands of characters, glyphs, and symbols	Specialty Variables	**Will only test on boolean and String
	true, false		
	Stores anything as a String literal between quotation marks ****This is not a primitive type****		

Declaring a Variable

- To declare a variable type, you identify your data type first, followed by a space, and then your identifier name:

Practice: Write the code on the line that declares the correct variable.

```
_____________________      //Declare number to be an integer variable

_____________________      //Declare gpa to be a double variable

_____________________      //Declare answer to be a boolean variable

_____________________      //Declare name to be a String variable
```

Assigning a Variable

- After you have a variable, you use it to _____
- To change the value of a variable, you use the assignment operator (_____)
- The variable name goes on the _____ of the equals sign and the expression (or value) goes on the _____ of the equal sign

Practice: Write the code on the line that assigns the value to the correct variable

```
_____________________ //Assign 5 to number
_____________________ //Assign 3.9 to gpa
_____________________ //Assign true to answer
_____________________ //Assign your name to name
```

Initializing a Variable

- Initializing a variable is when you declare a variable and assign it a value all on one line.

Practice: Write the code to initialize the variable to the correct value.

```
_____________________     //Initialize the integer variable number to 5

_____________________     //Initialize the double variable gpa to 3.9

_____________________     //Initialize the boolean variable answer to true

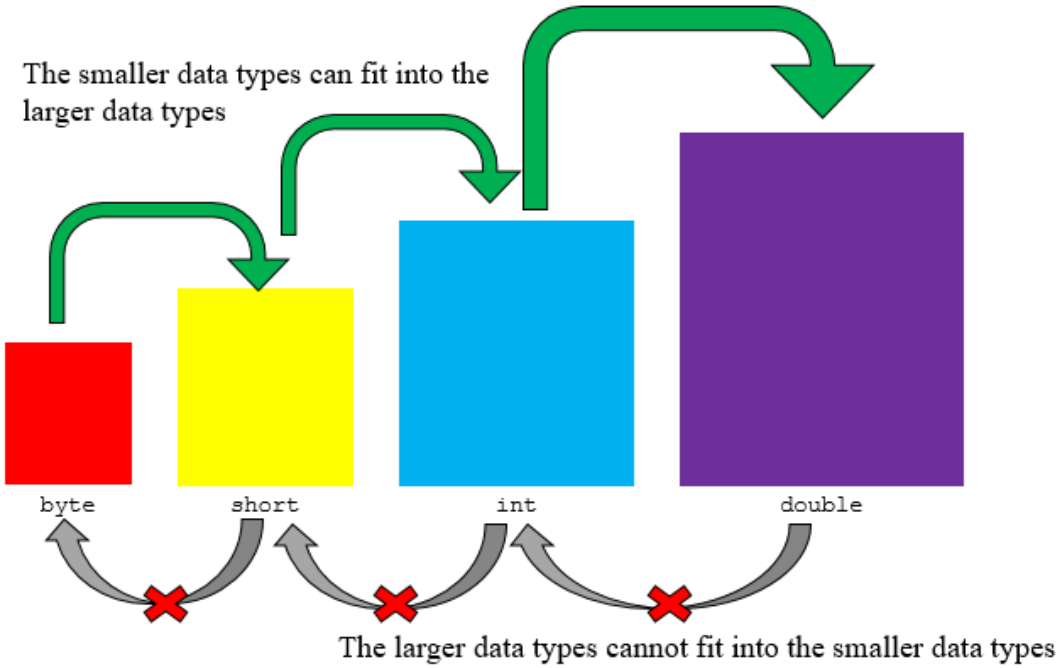
_____________________     //Initialize the String variable name to your name
```

Special initialization: If you have two or more variables of the same type, you can write the data type once, and then assign the values to each variable in the same statement:

```
int number = 5, otherNum = 6;
String name = "Kayla", middleName = "Jessica", lastName = "Smith";
```

Size Limits

- Each data type has a size limit, and if you try and store a value over that limit, you will get an error.
- We can mix data types, but you have to observe these size limits.



		Legal	Illegal
int vs double	• A double data type is larger than an integer data type • A double type _____ hold an integer • An integer _____ hold a double type	`double value = 4;` `double number = 4.0;`	`int otherValue = 4.5;` `int anotherOne = 4.0;`

String vs primitives	• A String can hold any characters
	• They MUST be enclosed in quotation marks
	• You can make a new String using two Strings put together with the + symbol
	• This is called concatenation
	• If you mix a String with a primitive, it will result in a String
		`String a = "5";`
		`String b = "Example" + 5;`
		`String c = "Example" + true;`
		`String d = true + "Example";`
			`String x = 5;`
			`String y = true;`